EHR Interface Specification

Specimen rejection notice (ORU^R01, outbound) and blood product dispense request (OMB^O27, inbound)

Item	Value
Document	Draft for review by the EHR interface team
System	Greyhound Blood Bank LIS (sending/receiving application **BloodBankLIS**, facility **BBLIS** by default)
HL7 version	2.5, ER7 encoding, MLLP transport
Version / date	0.1 / 2026-11-06
Decisions	OCD-062 (rejection notice) and OCD-063 (dispense request), decided by the product owner on 2026-11-06

Purpose. This document describes two messages the blood bank LIS can exchange with the EHR, so the EHR team can confirm they match what the EHR sends and accepts, or tell us what to change. Field positions are configurable in the LIS per interface, so most differences can be resolved by mapping rather than code changes.

What we need from you is in section 5: a short list of yes/no questions and two sample messages.

Current state. Both messages are built and tested in the LIS but are not in use. The rejection notice is only sent when an outbound results interface is enabled, and dispense requests are only acted on when the EHR sends one. Nothing changes for either system until both sides agree.

1. Conventions common to both messages

Topic	Behavior
Transport	MLLP over TCP. Host, port, and acknowledgement timeout are configured per interface (default timeout 15 seconds).
MSH-3 to MSH-6	Configurable per interface. Defaults: sending BloodBankLIS / BBLIS, receiving EHR / HOSP.
MSH-7 timestamps	yyyyMMddHHmmss in **UTC**, no offset. See question Q9.
MSH-11 / MSH-12	P (production) / 2.5
Acknowledgements	Original mode. MSA-1 AA = accepted. AE or AR = refused; MSA-3 carries the reason in plain language.
Codes	Test codes are the LIS's local codes (for example ABSC, ABORH, DAT) unless a value translation is configured in the LIS to your codes. See question Q2.
Errors	Every message is logged. Refused or unanswered messages go to the LIS interface error queue, where staff resolve, replay, or close them with a reason.

2. Specimen rejection notice (LIS to EHR, ORU^R01)

When it is sent. When the blood bank rejects a specimen (for example hemolyzed, clotted, or mislabeled), the LIS sends one message for each **open test** on that tube. A test is open when its order is not completed, cancelled, or discontinued, the line is active, and its result is not already verified. Product lines are not included.

What it means. "No result will be available on this specimen; a new specimen is needed." The order is **not cancelled**: it stays open, and when a new tube is collected and tested, the normal final result (OBX-11 F) follows for the same order.

Fields sent

Field	Content	Notes
MSH-9 / MSH-10	ORU^R01 / REJ{timestamp}{specimen id}-{n}	Unique per message.

Field	Content	Notes
PID-3	Patient MRN	LIS MRN. Assigning authority not sent today (Q8).
PID-5, PID-7, PID-8	Last^First, date of birth (yyyyMMdd), sex (M/F/O/U)	
OBR-1	1	
OBR-2	Placer order number	The order number the EHR sent with the order.
OBR-3	Filler order number, or the specimen accession	Accession when no filler number is stored.
OBR-4	Test code	Local code unless translated (Q2).
OBR-25	**X**	HL7 table 0123: no results available; order cannot be completed on this specimen.
OBX-1 / OBX-2	1 / TX	
OBX-3	Test code	Same as OBR-4.
OBX-5	Specimen {accession} rejected: {reason}. Recollect.	Reason is the rejection reason the technologist chose, with any comment.
OBX-11	**X**	HL7 table 0085: results cannot be obtained for this observation.

Example (one open antibody screen on tube ACC0001)

```
MSH|^~\&|BloodBankLIS|BBLIS|EHR|HOSP|20261106143000||ORU^R01|REJ20261106143000000123-1|P|2.5
PID|1||MRN0001||Demo^Patricia||19700101|F
OBR|1|ORD0001|ACC0001|ABSC||||||||||||||||||||X
OBX|1|TX|ABSC||Specimen ACC0001 rejected: Hemolyzed. Recollect.||||||X
```

Expected acknowledgement. MSA-1 AA. If the EHR returns AE or AR, or no acknowledgement arrives within the timeout, the message goes to the LIS error queue and is retried up to 5 times (configurable). If the LIS cannot queue the message at all, the rejection still stands and the technologist is told to phone the ward.

3. Blood product dispense request (EHR to LIS, OMB^O27)

Purpose. Lets the ward ask the blood bank, from the EHR, to send a unit that is **already reserved** for the patient, and say where to deliver it. In the LIS it becomes the same request a nurse can make in the LIS today: the unit moves to the top of the blood bank's Ready to issue list with the destination filled in.

Safety. The request never issues a unit. A technologist still issues it with every check the LIS applies (patient identification, unit scan, compatibility, expiration, and so on). The request cannot reserve a unit; it only asks for one that is reserved.

Fields read (defaults; each can be remapped per interface)

Data item	Default field	Use
Patient MRN	PID-3.1	Finds the patient. A merged chart is followed to the surviving chart.
Order control	ORC-1	NW or XO = request. CA, OC, or DC = withdraw an open request.
Placer order number	ORC-2.1	Identifies the unit when no DIN is sent: the order must have exactly one unit reserved for this patient.
Deliver to	BPO-14.1	Requested dispense-to location (for example OR 3).
Patient location	PV1-3.1	Used as the destination when BPO-14 is empty.
Unit (DIN)	*none*	Not part of a standard OMB. If your EHR can name the unit, tell us the field and we map it (Q6).
Requested by	ORC-12 (XCN)	Shown to the blood bank as "EHR (HL7): Last, First" and kept in the audit trail.

Data item	Default field	Use
Message control id	MSH-10	Recorded in the audit trail. A repeated control id is acknowledged without being applied again.

How the LIS decides

- **DIN sent** (if mapped): that unit must be reserved for this patient. It is used even when the order has several reserved units.
- **No DIN**: the placer order (ORC-2) must belong to this patient and have **exactly one** reserved unit.
- **Destination**: BPO-14, otherwise PV1-3. A request with neither is refused.
- **Same destination again**: accepted without change (safe to resend).
- **Different destination while a request is open**: refused. Send a cancel (ORC-1 CA) first, then the new request.
- **Cancel with nothing open**: accepted with "nothing to cancel".

Situation	ACK	MSA-3 text (example)
Request applied	AA	Issue of W0001230009201 requested to OR 3.
Cancel applied	AA	Issue request for W0001230009201 cancelled.
Patient not found	AE	No patient with MRN MRN0099.
Order has several reserved units, no DIN	AE	Order ORD0022 has 2 reserved units and the request names none; request the unit in the LIS or send its DIN.
No reserved unit on the order	AE	No unit is reserved on order ORD0022.
Order not on this patient	AE	Order ORD0099 is not on this patient.
No destination	AE	The request for W0001230009201 names no destination (BPO-14 or PV1-3).
Different destination already requested	AE	W0001230009201 was already requested at ... for ICU 4; cancel that request before asking for OR 3.

Every AE is also placed on the LIS interface error queue for a blood bank person to follow up.

Example request

```
MSH|^~\&|EHR|HOSP|BloodBankLIS|BBLIS|20261106143500||OMB^O27|EHR000101|P|2.5
PID|1||MRN0022^^^HOSP^MR||Demo-Multi^Riley||19580101|U
PV1||I|ICU 4
ORC|NW|ORD0022||||||||||1234^Wells^Dana
BPO|1|RBC^Red Blood Cells||||||||||||OR 3
```

Read by the LIS as: MRN0022, order ORD0022, deliver to OR 3, requested by Wells, Dana.

Example cancel (destination falls back to PV1-3, which a cancel does not need)

```
MSH|^~\&|EHR|HOSP|BloodBankLIS|BBLIS|20261106143500||OMB^O27|EHR000102|P|2.5
PID|1||MRN0022^^^HOSP^MR||Demo-Multi^Riley||19580101|U
PV1||I|ICU 4
ORC|CA|ORD0022||||||||||1234^Wells^Dana
BPO|1|RBC^Red Blood Cells||||||||||||
```

Example acknowledgements

```
MSH|^~\&|BloodBankLIS|BBLIS|EHR|HOSP|20261106143500||ACK|20261106143500123|P|2.5
MSA|AA|EHR000101|Issue of W0001230009201 requested to OR 3.

MSH|^~\&|BloodBankLIS|BBLIS|EHR|HOSP|20261106143500||ACK|20261106143500124|P|2.5
MSA|AE|EHR000101|Order ORD0022 has 2 reserved units and the request names none; request the unit in the LIS
or send its DIN.
```

4. Related messages already in place

- **BPS^O29 (LIS to EHR)** tells the EHR when a unit is dispensed or returned, so an accepted request is normally followed by a dispense status once the unit is issued.
- **RAS^O17 / BPS (EHR to LIS)** documents bedside administration. Unchanged by this document.
- **ORU^R01 corrections (OBX-11 C) and withdrawals (OBX-11 W)** are already sent automatically when a results interface is enabled.

5. Questions for the EHR team

#	Question	Answer
Q1	Does the EHR file an ORU with OBR-25 = X and OBX-11 = X, and where does the ward see it (results view, order status, in-basket)?	
Q2	Should test codes be sent as your codes? If so, please send the code list for ABORH, ABSC, DAT, and crossmatch so we can map them.	
Q3	Do you want one rejection message per test (current), or one per order with an OBX per test?	
Q4	Do you also want the order status changed (for example an ORC with order status), or is the result notice enough?	
Q5	Which message does the EHR send for a "send this unit now" request? If it is not OMB^O27, please send its specification.	
Q6	Can the request name the unit (DIN)? If so, in which field?	
Q7	Which field carries the deliver-to location: BPO-14, PV1-3, or another?	
Q8	Does the EHR need an assigning authority in PID-3 (for example MRN0001^^^HOSP^MR)?	
Q9	Should timestamps be sent in UTC (current) or local time with an offset?	
Q10	Can we test both messages in your test environment? Please give the test host, port, and a contact.	

Also helpful: one de-identified sample of each message as your system would send or expect it.

6. Next steps

- EHR team answers section 5 and sends samples.
- LIS team adjusts field mapping or code translation if needed (usually configuration, not code).
- Joint test in the EHR test environment: reject test tubes and confirm the ward display; send test dispense requests and confirm the blood bank list.
- Go-live decision for each message separately.